

Predicting metabolic rate during level and uphill outdoor walking using a low-cost GPS receiver

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de Müllenheim PY, Dumond R, Gernigon M, Mahé G, Lavenu A, Bickert S, Prioux J, Noury-Desvaux B, Le Faucheur A. Predicting metabolic rate during level and uphill outdoor walking using a low-cost GPS receiver. *J Appl Physiol* 121: 577–588, 2016. First published July 8, 2016; doi:10.1152/jappphysiol.00224.2016.—The objective of this study was to assess the accuracy of using speed and grade data obtained from a low-cost global positioning system (GPS) receiver to estimate metabolic rate (MR) during level and uphill outdoor walking. Thirty young, healthy adults performed randomized outdoor walking for 6-min periods at 2.0, 3.5, and 5.0 km/h and on three different grades: 1) level walking, 2) uphill walking on a 3.7% mean grade, and 3) uphill walking on a 10.8% mean grade. The reference MR [metabolic equivalents (METs) and oxygen uptake ($\dot{V}O_2$)] values were obtained using a portable metabolic system. The speed and grade were obtained using a low-cost GPS receiver (1-Hz recording). The GPS grade (Δ altitude/distance walked) was calculated using both uncorrected GPS altitude data and GPS altitude data corrected with map projection software. The accuracy of predictions using reference speed and grade (actual_[SPEED/GRADE]) data was high [$R^2 = 0.85$, root-mean-square error (RMSE) = 0.68 MET]. The accuracy decreased when GPS speed and uncorrected grade (GPS_[UNCORRECTED]) data were used, although it remained substantial ($R^2 = 0.66$, RMSE = 1.00 MET). The accuracy was greatly improved when the GPS speed and corrected grade (GPS_[CORRECTED]) data were used ($R^2 = 0.82$, RMSE = 0.79 MET). Published predictive equations for walking MR were also cross-validated using actual or GPS speed and grade data when appropriate. The prediction accuracy was very close when either actual_[SPEED/GRADE] values or GPS_[CORRECTED] values (for level and uphill combined) or GPS speed values (for level walking only) were used. These results offer promising research and clinical applications related to the assessment of energy expenditure during free-living walking.

wearable activity monitors; energy expenditure; equations; global positioning system; grade

NEW & NOTEWORTHY

This is the first study to characterize the direct relationship between global positioning system (GPS) speed and grade and

the metabolic rate while walking outdoors under different controlled speed and grade conditions. Using GPS speed and grade yields accurate metabolic rate predictions during level and uphill outdoor walking, particularly when GPS grade is corrected. Moreover, when using GPS parameters with published speed/grade-based equations, the metabolic rate predictions were close to those obtained using actual speed and grade values.

WALKING IS A POPULAR AND ACCESSIBLE physical activity (15) that provides health benefits in the form of a reduced risk of mortality (19, 50–52). It has been recommended by scientific authorities for maintaining health (21) and for preventing and managing chronic diseases (16). Accurately predicting the walking metabolic rate (i.e., the physiological intensity of walking) is of importance in both health and disease. Indeed, because the walking metabolic rate is directly related to a given rate of energy expenditure (EE), it is an important determinant of the magnitude of the health benefits that walking provides (21) and is a triggering factor of most of the pathophysiological responses and adaptations to walking (e.g., during rehabilitation) in chronic disease patients (16).

The ability to predict the walking metabolic rate is particularly important in free-living conditions, in which the direct measurement of metabolic rate is often impractical. Commercially available wearable sensors can measure signals related to movement in free-living conditions (11). Such sensors can use existing equations to predict the metabolic rate based on the signals they measure. In the long tradition of studies that aim to develop prediction equations for the walking metabolic rate, the walking speed and grade have emerged as two powerful predictors (e.g., 8, 18, 28, 30, 35, 49). Thus wearable sensors that are capable of measuring speed and grade are potentially of great interest for predicting the metabolic rate during free-living walking, particularly during outdoor walking, given that people generally use outdoor settings such as parks and neighborhood streets for walking (15).

Low-cost global positioning system (GPS) receivers have been increasingly used over the last 10 yr to assess physical activity locations and the complex interaction between physical activity and the outdoor environment (25). Another major factor of interest regarding these low-cost GPS receivers is their accuracy in estimating speed during walking (26, 34).

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GPS receivers also allow the altitude to be measured over time; thus the grade of the terrain can be calculated by dividing the altitude variation by the distance traveled. Consequently, it can be assumed that GPS units are good candidates for estimating the metabolic rate during level and graded outdoor walking.

To the best of our knowledge, no study has systematically examined low-cost GPS receivers' accuracy for such estimations. Although a few studies have used GPS receivers to estimate EE during outdoor walking, none of these studies have included all the required methodological conditions (13, 14, 22, 33). Importantly, the experimental constraint of finding outdoor pathways that include both different grades and sufficient distance to achieve a steady state at different walking speeds can be highly challenging. This may explain why 1) no equation for predicting the metabolic rate during level and graded outdoor walking has been developed using data obtained exclusively in outdoor settings and 2) the literature does not report how the accuracy of the GPS technique for measuring speed and grade translates into a sufficiently accurate prediction of the metabolic rate.

The purpose of the present study was to assess the accuracy of using a low-cost GPS receiver to estimate the metabolic rate during level and uphill outdoor walking. To achieve this objective, we first developed and compared initial equations for predicting metabolic rate that were obtained using actual and GPS speed and grade parameters. Second, we cross-validated published predictive equations for the walking metabolic rate using speed or both speed and grade parameters. In this way, we aimed to investigate whether the precision of metabolic rate estimation was affected by the use of GPS speed and grade values.

METHODS

Study Design

The present study was a part of the Global Positioning Satellite and Accelerometry to Assess Human Locomotion (Acti-GPS) protocol approved by the institutional ethics committee of the University Hospital in Angers (Comité de Protection des Personnes: OUEST II, Angers, France). The protocol was registered with the American National Institutes of Health database under reference no. NCT01805219. Healthy young adult subjects were recruited to participate in the present study. As recommended for criterion-related validity studies (29), we aimed to include a minimum of 30 subjects [male:female (M/F) = 15:15]. Each subject signed an informed consent form and then underwent the following experimental procedure: 1) an inclusion visit, 2) peak oxygen uptake ($\dot{V}O_{2\text{ peak}}$) measurement during an incremental field running test, and 3) three outdoor walking sessions. There were at least 24 h between the incremental running test and the outdoor walking sessions. Furthermore, the three outdoor walking sessions were performed on two or three different days. For both the incremental field running test and the outdoor walking sessions, the subjects were instructed to refrain from vigorous physical activity and alcohol ingestion for at least 24 h prior to testing. In addition, the subjects were instructed to refrain from smoking and ingesting food and caffeine for at least 2 h prior to testing.

Experiments

Inclusion visit. The subjects underwent a medical history and a physical examination at the University Hospital to ensure good health. Weight, height, and body fat percentage were measured during the examination. A resting electrocardiogram was also performed. The body fat percentage was obtained using the skin fold method (2). Skin

folds were measured at three body sites (the chest, triceps, and subscapular region in the men and the triceps, abdomen, and suprailiac region in the women) using a Harpenden caliper (Baty, Burgess Hill, UK) according to the manufacturer's guidelines. Suitable equations for young men and women were used to determine the body fat percentage (2).

Peak oxygen uptake measurement. The $\dot{V}O_{2\text{ peak}}$ value was determined during an outdoor maximal incremental field running test. The subjects were equipped with a K4b² portable metabolic system (Cosmed, Rome, Italy) for gas exchange measurement. Thereafter, the subjects performed a modified version of the University of Montreal Track Test (27) on the first lane of a 400-m athletic track. The running speed started at 8.5 km/h and increased by 0.5 km/h every minute until exhaustion (10). The test ended when either the subject could no longer maintain the speed imposed by the beep or the subject voluntarily stopped.

Outdoor walking sessions. The subjects completed three outdoor walking sessions under three different conditions corresponding to three different grades, including 1) a level walking session (0.0%), 2) an uphill walking session on a 3.7% grade, and 3) an uphill walking session on a 10.8% grade. The level walking session took place on a 400-m athletic track. The uphill outdoor walking sessions on 3.7 and 10.8% grades were performed on two different hiking trails. At each location, the actual (reference) grade of the walking courses was measured every 25 m along the courses by a certified surveyor using a Trimble S6 Total Station (Trimble, Sunnyvale, CA) and a Trimble R8 Global Navigation Satellite System. Because of experimental constraints, nine subjects used a second athletic track that was located in another area but had topographic features identical to those of the first track (actual grade of 0.0%). The actual grade of each hiking trail differed slightly according to the walking speed tested. For the sake of readability, values of 3.7 and 10.8% for the two uphill walking sessions (hiking trails) are used throughout the text and correspond to the mean grade of each uphill walking session.

For each walking session, the subjects walked for three periods of 6 min each at 2.0, 3.5, and 5.0 km/h. The sequence of the three outdoor walking sessions and the three walking speeds within each session was randomized for each subject. The subjects were asked to wear the same shoes for all three walking sessions. For each walking session, the walking course was indicated by colored cones placed on the ground at regular intervals. The subjects regulated their walking speed using these colored cones and by listening to sounds on an MP3 player. The MP3 player emitted beeps at regular time intervals for the walking speed being tested. All of the walking sessions took place under the supervision of an investigator who 1) ensured that the subjects walked at the correct speed and 2) measured the actual walking time using a stopwatch. Then, the actual (reference) walking speed was calculated for each walking period by dividing the total distance walked by the time. For each walking session, the first walking period was preceded by a 5-min standing resting period (baseline). Between each walking period, the subjects were given a sufficient recovery period to allow their heart rate and $\dot{V}O_2$ to return to baseline values.

Instrumentation and Data Collection

Portable metabolic system. During all walking sessions, we used the Cosmed K4b² breath-by-breath portable metabolic system to measure gas exchange. Although the K4b² is not the gold standard for metabolic measurements, it provides valid (24) and reliable (12) measurements of gas exchange at different walking and running intensities. We tested the reliability of our K4b² system for the measurement of $\dot{V}O_2$ while walking on a treadmill on a 10% grade and at walking speeds between 2 and 6 km/h. We obtained the following results: coefficient of variation [95% confidence interval (CI)] = 6.2% (4.7; 9.0); intraclass correlation coefficient = 0.97 (0.94; 0.99); bias (95% CI) = -0.07 (-0.36; 0.22) l/min; 95%

limits of agreement = ± 0.29 l/min (unpublished results). The K4b² consists of an analyzer unit and a battery, which are fixed to the subject's chest and back, respectively, with a lightweight anatomical harness. Further details about the system are provided elsewhere (38). The K4b² was calibrated before the beginning of each walking session according to the manufacturer's guidelines.

GPS receiver. The subjects wore a DG100 GPS receiver (DG100; GlobalSat, Taipei, Taiwan) during all walking sessions. The DG100 (dimensions $80 \times 55 \times 18$ mm, weight ~ 200 g, data capacity 60,000 points, battery life ~ 17 h) is a low-cost GPS receiver (approximately US\$60) that includes the European Geostationary Navigation Overlay Service function. High accuracy and high reliability, as well as a low interunit variability, in the measurement of walking speeds and distances have been previously reported for the DG100 (1, 34). Moreover, the use of the DG100 has been shown to be applicable in a clinical population for the nonsupervised recording of walking ability in the community (17). In the present study, the unit was placed in a backpack, and its external antenna (AT-65 GPS Active Antenna; GlobalSat) was fixed on the subject's right shoulder (on the superior border of the right scapula). The DG100 was set at a 1-Hz sampling rate before each outdoor walking session using a personal computer and the manufacturer's software (Data Logger Utility, version 1.1).

Data collection. The DG100 was initialized ~ 10 min before the beginning of each outdoor walking session. At the beginning and end of each walking period, the current Coordinated Universal Time (www.timeanddate.com/worldclock) was reported to the nearest second, and the K4b² unit was simultaneously used to insert a time mark in the data file. This procedure ensured that during the subsequent data analysis, the gas exchange data for each walking period could be matched with the data recorded by the DG100. At the end of each outdoor walking session, the data were downloaded from the K4b² (K4b² Data Management Software, version 10.0 e) and the DG100 (Data logger PC utility, version 1.1, 2006) via a personal computer for subsequent analysis.

Data Analysis

Gas exchange. Downloaded gas exchange data were exported to an Excel spreadsheet (version 2010) for analysis. The $\dot{V}O_{2\text{peak}}$ was recorded as the highest 30-s average $\dot{V}O_2$ value measured during the maximal incremental field running test. To ensure that the test value was maximal and to validate the obtained $\dot{V}O_{2\text{peak}}$ value, the subjects were required to have a peak respiratory exchange ratio ≥ 1.10 and a peak heart rate $\geq 95\%$ of the age-predicted maximum (44).

The absolute $\dot{V}O_2$ values obtained from the different walking sessions were averaged over the last 3 min of each walking period. Because the terrains differed among the three walking conditions, we adjusted the average $\dot{V}O_2$ values using terrain coefficients (η) for energy cost prediction (42). A terrain coefficient indicates how the energy cost of walking is modified during walking on a given surface compared with walking on a reference surface. In the original study by Soule and Goldman (42), the reference surface was the treadmill, and the energy cost of walking on a blacktop surface was shown to be similar to the energy cost of walking on the treadmill ($\eta = 1.0$). In the present study, for the level outdoor walking sessions, we assumed that the effect of the terrain on the walking cost did not differ between the two athletic tracks. We considered the walking cost for these two tracks to be similar to the walking cost of a blacktop surface. The two hiking trails used for the uphill outdoor walking sessions were near-dirt roads, which are 1.10 times more costly than a blacktop surface (42). The hiking trail for the uphill outdoor walking sessions on a 3.7% grade was composed of packed earth, which was assumed to be slightly less costly than a dirt road. The hiking trail for the uphill outdoor walking sessions on a 10.8% grade was characterized by packed earth and stony ground and thus was considered slightly more costly than a dirt road. Thus we defined terrain coefficients of $\eta = 1.08$ and $\eta = 1.12$ for the uphill outdoor walking sessions at the

grades of 3.7 and 10.8%, respectively (42). Accordingly, the average absolute $\dot{V}O_2$ values that were calculated for walking on the hiking trails with a 3.7% and a 10.8% grade were divided by the values of the corresponding coefficients (i.e., by 1.08 for the $\dot{V}O_2$ values calculated for walking on a 3.7% grade and by 1.12 for the $\dot{V}O_2$ values calculated for walking on a 10.8% grade). Then, the absolute $\dot{V}O_2$ values were converted to relative $\dot{V}O_2$ values ($\text{ml}\cdot\text{min}^{-1}\cdot\text{kg}^{-1}$) by dividing the absolute $\dot{V}O_2$ by the subject's body mass plus the equipment mass. Metabolic equivalent (MET) values were obtained by dividing the relative $\dot{V}O_2$ values by 3.5 (1 MET = $3.5 \text{ ml}\cdot\text{min}^{-1}\cdot\text{kg}^{-1}$).

GPS data. To facilitate the identification of each walking period in the GPS speed data file, we used the time stamp that the investigator provided during the walking sessions along with a previously validated methodology for the detection of walking bouts using the GPS speed (26). For each walking period, the GPS speed data were averaged over the last 3 min to match the average $\dot{V}O_2$ data. Furthermore, for each walking period, the corresponding mean GPS grade was obtained by averaging the GPS grade values calculated over time intervals of 18 s. In this manner, regardless of the walking speed, the mean GPS grade was always computed using 10 values ($10 \times 18 \text{ s} = 180 \text{ s} = 3 \text{ min}$). The GPS grade over each 18-s time interval was calculated as follows: $[\text{altitude variation (m)} / \text{traveled distance (m)}] \times 100$. This approach was used in two ways to compute the uncorrected and corrected GPS grades. First, raw altitude GPS data were used to obtain uncorrected GPS grade values. Second, to obtain corrected GPS grade values, we uploaded the GPS files to map projection software (CartoExploreur, version 3.11.0, build 2.6.6.22; Bayo, Ap-poigny, France). Using the French National Institute of Geography map projections that the software provided, each GPS altitude value could be corrected for each corresponding GPS position value for latitude and longitude (a more detailed description of this procedure is available on simple request to the authors).

Statistical Analysis

Descriptive statistics and assessment of the accuracy of GPS for estimating speed/grade. The data are reported as means $\pm$ SD or medians with interquartile range (IQR = 25th to 75th percentiles) when appropriate. To assess the accuracy of the GPS receiver method for estimating walking speed and grade, we computed the typical error of the estimate (TEE) and the coefficient of variation (CV, for the GPS accuracy of speed estimation only) with 95% confidence intervals (95% CI) according to a modified procedure (34) of the method originally proposed by Hopkins (23).

Linear mixed models and leave-one-out validation for predicting the metabolic rate. To assess the accuracy obtained using actual and GPS speed and grade parameters for predicting the metabolic rate (METs and $\dot{V}O_2$), we developed prediction equations using 1) both the actual speed and grade, 2) both the GPS speed and the uncorrected grade, and 3) both the GPS speed and the corrected grade. These equations were developed for 1) the combined level and uphill outdoor walking conditions and 2) the level outdoor walking condition only. The latter models were also developed because most of the published prediction equations for the walking metabolic rate were developed during level walking. Consistent with conventional equations for predicting the metabolic rate during level and graded walking (2), we used a three-component model that included the following: 1) the O_2 requirement related to resting metabolism, described by the value of the intercept given by the statistical model, which was not required to be the standard value of resting metabolic cost as in the American College of Sports Medicine's equation (2); 2) the O_2 requirement of the horizontal component of the walk, namely, speed (V , km/h) $\times$ a coefficient provided by the statistical model; 3) the O_2 requirement of the vertical component of the walk, namely, $V \times$ percentage of grade (G , %) $\times$ a coefficient provided by the statistical model (models without the grade component were also developed for the sole level walking condition).

Table 1. Prediction equations from literature for walking metabolic rate ($\text{ml O}_2 \cdot \text{min}^{-1} \cdot \text{kg}^{-1}$)

	<i>n</i>	Equation	Condition	Intensity Range ^a	
				<i>V</i>	<i>G</i>
Level and uphill walking					
ACSM (2)	–	$3.5 + 0.1 \cdot (V/0.06) + 1.8 \cdot (V/0.06) \cdot (G/100)$	TM	–	–
Pandolf et al. (35) ^b	6	$\{1.5 \cdot BM + 2.0 \cdot (BM + L) \cdot (L/BM)^2 + \eta \cdot (BM + L) \cdot [1.5 \cdot (V/3.6)^2 + 0.35 \cdot (V/3.6) \cdot G]\} \times [(0.01433/4.887 \times 1,000)/(BM + L)]$	TM	0.7–8.5	0–24
Level walking					
Blessey et al. (6)	40	$7.55 + 0.000811 \cdot (V/0.06)^2$	OG	3.2–7.3	NA
Pearce et al. (36)	20	$6.28 - 1.81 \cdot (V/3.6) + 4.38 \cdot (V/3.6)^2$	OG	2.3–12.0	NA
Brooks et al. (8)	72	$(1.034 - 0.016 \cdot M - 0.196 \cdot Gen + 0.832 \cdot V) \times 3.5$	OG	4.0–6.5	NA
Van der Walt and Wyndham (47)	6	$0.00599 \cdot M + 0.000366 \cdot M \cdot V^2$	TM	3.2–8.0	NA
Bubb et al. (9)	23	$21.11 - 0.3593 \cdot (V/0.06) + 0.0030 \cdot (V/0.06)^2$	TM	4.8–7.8	NA
Balogun et al. (4)	25	$17.77 - 0.3023 \cdot (V/0.06) + 0.0027 \cdot (V/0.06)^2$	TM	3.2–7.8	NA
Ludlow and Weyand _[GENERAL] ^c (28)	–	$RMR + 3.90 + 6.05 \cdot (V/3.6)^2 \cdot Ht^{-1}$	TM	1.4–6.8	NA
Ludlow and Weyand _[DOUGLAS] ^d (28)	–	$RMR + 3.85 + 5.97 \cdot (V/3.6)^2 \cdot Ht^{-1}$	TM	1.6–6.5	NA
Ludlow and Weyand _[ADULTS] ^e (28)	–	$7.15 + 5.97 \cdot (V/3.6)^2 \cdot Ht^{-1}$	TM	1.6–6.5	NA

Here, *n*, sample size; –, information not available; *V*, speed (km/h); *G*, grade (%); *BM*, body mass (kg); *L*, load carried (kg); η , terrain factor ($\eta = 1.0$ for the treadmill or the blacktop surface); *M*, subject's mass (the cross-validation was performed using the subject's total mass, i.e., using the body mass plus the equipment mass); *Gen*, gender (1, male; 2, female); *RMR*, resting metabolic rate ($\text{ml O}_2 \cdot \text{min}^{-1} \cdot \text{kg}^{-1}$); *Ht*, height (m); TM, treadmill walking; OG, over-ground walking; NA, not applicable. ^aIntensity range over which the equation was validated. ^bThe coefficient $[(0.01433/4.887 \times 1000)/(M + L)]$ was used to convert watts (i.e., units used by authors' formula) to $\text{ml O}_2 \cdot \text{min}^{-1} \cdot \text{kg}^{-1}$. ^cEquation developed on 127 $\dot{V}\text{O}_2$ group means (acquired using indirect calorimetry) retrieved from literature. ^dEquation developed on 42 $\dot{V}\text{O}_2$ group means (acquired using Douglas bags method only) retrieved from literature. ^eEquation derived from the Ludlow and Weyand_[DOUGLAS] equation, for adults only.

A linear mixed model approach was used to account for the dependence among repeated measurements obtained from the same subject, as previously noted (48). Assuming that y_{ij} represents the response for the *j*th observation of the *i*th participant, the model that combines the level and uphill outdoor walking conditions can be written as follows:

$$y_{ij} = \alpha + \beta \cdot V_{ij} + \gamma \cdot V_{ij} \cdot G_{ij} + a_i + b_i \cdot V_{ij} + c_i \cdot V_{ij} \cdot G_{ij} + e_{ij}$$

where α , β , and γ are fixed effects that denote the population's average intercept, the average speed for the parameter *V*, and the average grade for the parameter *G*, respectively. The terms a_i , b_i , and c_i represent each subject's random intercept and slope deviations around the corresponding population parameters, and e_{ij} represents the random error. Prediction equations were developed using linear mixed models for the metabolic rate ($\dot{V}\text{O}_2$ and MET) and for *I*) both the actual speed and grade and 2) both the GPS speed and grade, using the GPS grade computed with either uncorrected or 3) corrected altitude data. The mass, gender, and body fat percentage were added as covariates via a forward stepwise selection procedure.

Each model was fitted using the restricted maximum likelihood, the standard method used to estimate variance components in linear mixed models (37). The parameter estimates (fixed-effects coefficients and variance components) were obtained using a covariance structure chosen to optimize goodness of fit: diagonal for random subject effects and compound-symmetrical for random errors. The R^2_m (the term "m" indicates a marginal R^2 ; that is, the variance is explained by fixed factors) was computed according to the method proposed by Nakagawa and Schielzeth (32). The standard error of the estimate (SEE) and Akaike information criteria (46) were reported for each model. The smaller the value of the Akaike information criteria (AIC), the better the predictive model was.

The accuracy of the linear mixed models developed in the present study was then tested using the leave-one-out method, as previously proposed (43). The leave-one-out cross-validation approach is a preferred alternative to the "split sample" method (43). The level of accuracy reached with each model was reported as the average root-mean-square error (RMSE): the lower the RMSE, the higher the accuracy. Statistical modeling was performed using R software version 3.2.1 (39a). In this paper, only the fixed-effects coefficients and the corresponding statistics are reported.

Cross-validation of previously published equations. We cross-validated several published prediction equations for the walking metabolic rate (Table 1) using either the actual or GPS speed and grade parameters. We used published prediction equations developed for both level and uphill walking and also for level walking only. The American College of Sports Medicine (ACSM) (2) and Pandolf et al. (35) equations were also cross-validated using data from the level walking condition only. In this case, only speed was used. The predicted $\dot{V}\text{O}_2$ values obtained for the uphill outdoor walking periods using the ACSM equation were adjusted for the energy cost of the terrain using the corresponding terrain coefficients previously described (refer to DATA ANALYSIS). For the Pandolf et al. equation, our terrain coefficients were directly incorporated within the equation (Table 1). For each cross-validated equation, we computed the bias, the standard deviation of the differences between the measured (without adjustment for the energy cost of the terrain) and predicted metabolic rate values, the 95% limits of agreement (95% LoA) using R software, and the RMSE using Excel spreadsheets. For each cross-validated equation, a Friedman test was performed to determine whether some differences existed between the measured values of metabolic rate ($\dot{V}\text{O}_2$) and the predicted values of metabolic rate. Predicted metabolic rate was computed either using the actual parameters of speed and/or grade or using the GPS parameters of speed and/or grade (using uncorrected and corrected grade values for the level and uphill walking condition). Because the data did not follow a normal distribution, the Friedman test was used. When the Friedman test result was significant, a post hoc Wilcoxon signed ranks test was performed for pairwise comparisons with Bonferroni correction. The level of statistical significance was fixed at $P < 0.05$. The Friedman and Wilcoxon tests were performed using the Statistical Package for the Social Sciences, version 23.0 (SPSS, Chicago, IL).

RESULTS

Thirty-one subjects were included in the present study. One subject was excluded because she began taking medication during the study. Thus a total of 30 healthy subjects were analyzed in the present study (F/M = 15:15). The mean $\pm$ SD values for age, height, body mass, body fat, and $\dot{V}\text{O}_{2\text{peak}}$ were 22.0 ± 1.4 yr, 172 ± 6 cm, 67.1 ± 9.3 kg, $14.0 \pm 5.4\%$, and

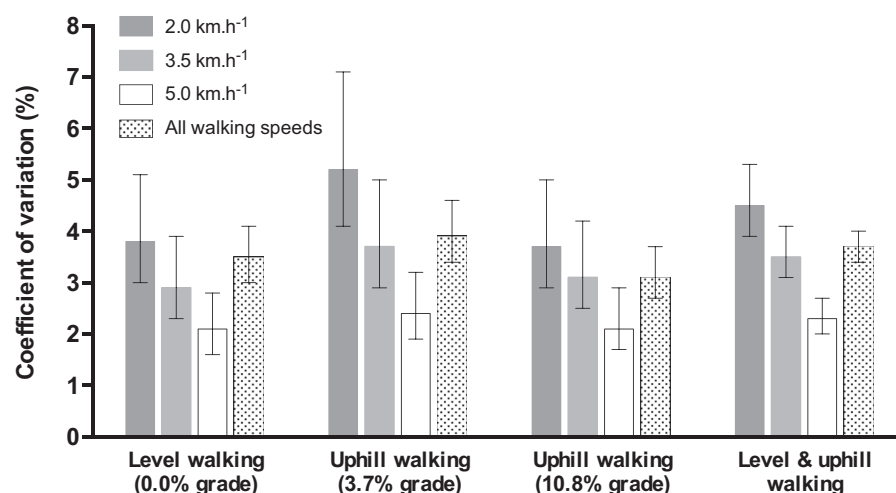


Fig. 1. Accuracy of the DG100 GPS for estimating the mean walking speed during the level and uphill outdoor walking sessions. Bars indicate 95% confidence intervals.

$52.1 \pm 7.6 \text{ ml O}_2 \cdot \text{min}^{-1} \cdot \text{kg}^{-1}$, respectively. One subject could not perform the maximal incremental running field test because of a schedule constraint. All of the GPS data were correctly recorded except for one subject's data for the uphill outdoor walking session on a 3.7% grade. The GPS data for the corresponding three walking periods were not recorded by the GPS receiver, likely because of a loss of the GPS satellite signals. Thus a total of 267 walking periods were available for the statistical modeling.

GPS Accuracy for Estimating Speed and Grade

The TEE (95% CI) for speed estimation was 0.1 (0.1–0.1) km/h for most of the outdoor walking sessions, regardless of the walking speed. Figure 1 shows the CVs for GPS accuracy for speed estimation. The CVs ranged from 2.1 to 5.2% among the different walking sessions. Importantly, for each walking condition, the CVs decreased with increasing walking speed. Figure 2 shows the TEE (95% CI) of the GPS receiver for estimating the mean grade during the outdoor walking sessions using the uncorrected (Fig. 2A) and corrected (Fig. 2B) altitude data. The TEE (95% CI) values for the walking sessions as a whole were 3.4 (3.1–3.7)% and 0.9 (0.9–1.0)% using uncorrected and corrected altitude data, respectively. For each walking condition, the GPS error in grade estimation decreased with increased walking speed for both the uncorrected and corrected GPS data.

GPS Accuracy for Predicting Metabolic Rate

Figure 3 shows the metabolic rate data measured during the level and graded outdoor walking sessions, expressed in METs (Fig. 3A) and in $\dot{V}\text{O}_2$ (Fig. 3B). Medians increased with walking speed, and this increase was larger when the grade was steeper. Table 2 displays the results of the different statistical models that were computed using either the actual values or the GPS parameters for speed and grade (when appropriate). Regarding the models obtained for the combined level and uphill outdoor walking conditions, the actual speed and grade accounted for >80% of the metabolic rate variance. The average RMSEs for these models were below 0.70 METs and $2.50 \text{ ml O}_2 \cdot \text{min}^{-1} \cdot \text{kg}^{-1}$. When using GPS speed and GPS uncorrected grade parameters, the variance that the statistical models explained decreased to below 70%, and the average RMSEs increased to 1.00 MET and $3.49 \text{ ml O}_2 \cdot \text{min}^{-1} \cdot \text{kg}^{-1}$. However, compared with the GPS speed and GPS uncorrected grade, the combination of GPS speed and corrected GPS grade values increased the variance explained by the statistical models (to over 80%), with average RMSEs below 0.80 METs and $2.80 \text{ ml O}_2 \cdot \text{min}^{-1} \cdot \text{kg}^{-1}$. On the basis of the obtained AIC and SEE values, the models obtained similar rankings (Table 2). Adding mass, gender, and body fat percentage to the models did not improve the goodness of fit (greater AIC values; $P > 0.05$). As shown in Table 2, the models computed for the level walking

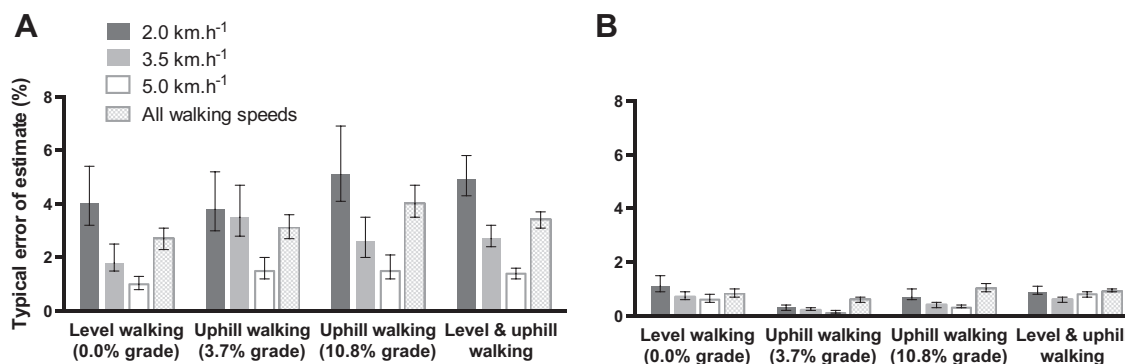


Fig. 2. Accuracy of the DG100 GPS for estimating the mean grade (%) using the uncorrected altitude data (A) and using the corrected altitude data (B) during the level and uphill outdoor walking sessions. Bars indicate 95% confidence intervals. The corrected altitude data were obtained using map projection software.

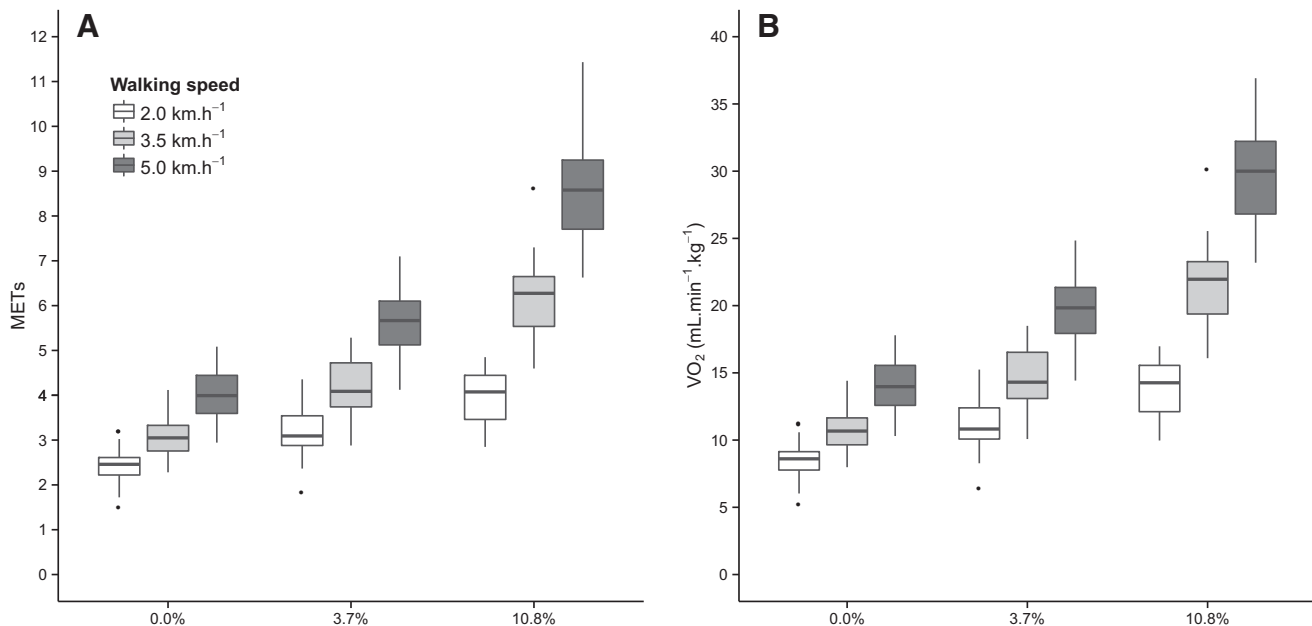


Fig. 3. Boxes and whiskers illustrating the individual means of the metabolic rate in METs (A) and in $\dot{V}O_2$ (B) obtained during the last 3 min of the walking periods at the different walking speeds and grades tested. The metabolic rate values displayed here are the measured values that have been adjusted for the energy cost of the terrain. The x-axis values are the means of the grade values corresponding to the walking periods tested in a given location. Each box represents the interquartile range (IQR) of the individual mean values, with the lower and upper hinges representing the 25th and 75th percentiles, respectively. The lower whiskers represent the smallest observations that were greater than or equal to the lower hinges $-1.5 \times \text{IQR}$. The upper whiskers represent the highest observations that were lower than or equal to the upper hinges $+1.5 \times \text{IQR}$. Points are outliers (i.e., values that were lower or greater than the smallest or greatest values corresponding to the whiskers' extremities). METs, metabolic equivalents; $\dot{V}O_2$, oxygen uptake.

condition presented a similar accuracy regardless of whether the actual or GPS parameters were used ($R^2 = \sim 65\%$, $\text{RMSE} = \sim 0.45 \text{ METs}/\sim 1.50 \text{ ml O}_2\cdot\text{min}^{-1}\cdot\text{kg}^{-1}$).

Cross-validation of the Published Equations for Predicting Metabolic Rate

Tables 3 and 4 present the cross-validation results for the published prediction equations for our measured $\dot{V}O_2$ values. Regarding the metabolic rate predictions for the combined level and uphill outdoor walking conditions (Table 3), the RMSE was between 3.3 and 4.0 ml $\text{O}_2\cdot\text{min}^{-1}\cdot\text{kg}^{-1}$ for both equations tested when using the actual speed and grade. The RMSE increased ($>4.5 \text{ ml O}_2\cdot\text{min}^{-1}\cdot\text{kg}^{-1}$) when the GPS speed and uncorrected grade were used, but decreased ($\leq 4.1 \text{ ml O}_2\cdot\text{min}^{-1}\cdot\text{kg}^{-1}$) when using the GPS speed and corrected grade. Bias, SD, and 95% LoA tended to follow the same trends as the RMSE. Importantly, the accuracy of the prediction obtained using the actual speed and grade and using the GPS speed and corrected grade was very similar. Although the ACSM (2) and Pandolf et al. (35) equations allowed similar RMSEs to be obtained when using GPS speed and corrected grade, the bias obtained using the ACSM equation was lower than that achieved with the Pandolf et al. equation. Furthermore, for the ACSM equation, when compared with the measured metabolic rates [median (IQR): 15.3 (11.3–21.5) ml $\text{O}_2\cdot\text{min}^{-1}\cdot\text{kg}^{-1}$], the metabolic rates estimated with the actual speed and grade [14.3 (9.7–18.9) ml $\text{O}_2\cdot\text{min}^{-1}\cdot\text{kg}^{-1}$], the GPS speed and uncorrected grade [13.1 (9.8–20.2) ml $\text{O}_2\cdot\text{min}^{-1}\cdot\text{kg}^{-1}$], or the GPS speed and corrected grade [15.1 (10.4–19.0) ml $\text{O}_2\cdot\text{min}^{-1}\cdot\text{kg}^{-1}$], were significantly lower ($P < 0.05$). When

the Pandolf et al. equation was used, the estimated metabolic rates were significantly lower than the measured values ($P < 0.05$) regardless of which speed and grade parameters (actual or GPS values) were used. However, for the Pandolf et al. equation, the metabolic rate estimated using the GPS speed and corrected grade [13.0 (8.7–19.3) ml $\text{O}_2\cdot\text{min}^{-1}\cdot\text{kg}^{-1}$] was significantly higher than the metabolic rate estimated using the actual speed and grade [12.7 (8.4–19.2) ml $\text{O}_2\cdot\text{min}^{-1}\cdot\text{kg}^{-1}$].

The RMSEs related to the metabolic rate predictions for the level outdoor walking condition are shown in Table 4. All of the RMSEs obtained using either the actual speed or the GPS speed were between 1.70 and 3.25 ml $\text{O}_2\cdot\text{min}^{-1}\cdot\text{kg}^{-1}$. The two most established and commonly used predictive equations, the ACSM (2) and Pandolf et al. (35) equations, tended to underestimate the measured $\dot{V}O_2$ values. The Ludlow and Weyand_[ADULTS] (28) equation seemed to yield the most accurate results in our population (Table 4). Of interest, the RMSEs obtained using either the actual speed or the GPS speed were nearly the same for all of the equations. Moreover, for all but one equation, there was no significant difference when comparing the metabolic rates estimated using the actual speed to those estimated using the GPS speed (Table 5). Additionally, for some of the equations used, the metabolic rates estimated using the GPS speed were not significantly different from the measured metabolic rate (Table 5).

Figures 4 and 5 graphically illustrate the variability in the relationship between the measured (without adjustment for the energy cost of the terrain) metabolic rate values and the predicted metabolic rate values obtained using previously published predictive equations and using either actual

Table 2. MET and $\dot{V}O_2$ prediction equations developed in the present study

Model	Equation	R^2	AIC	SEE	RMSE
<i>Level walking and uphill walking</i>					
MET prediction					
Actual[SPEED AND GRADE]	$1.1692 + 0.5728 \cdot V + 0.0841 \cdot V \cdot G$	0.85	439.73	0.47	0.68
GPS[UNCORRECTED]	$1.2556 + 0.7212 \cdot V + 0.0510 \cdot V \cdot G$	0.66	699.49	0.95	1.00
GPS[CORRECTED]	$0.6991 + 0.7141 \cdot V + 0.0809 \cdot V \cdot G$	0.82	600.59	0.63	0.79
$\dot{V}O_2$ (ml $O_2 \cdot \text{min}^{-1} \cdot \text{kg}^{-1}$) prediction					
Actual[SPEED AND GRADE]	$4.0472 + 2.0167 \cdot V + 0.2938 \cdot V \cdot G$	0.86	1,106.77	1.64	2.39
GPS[UNCORRECTED]	$4.3467 + 2.5355 \cdot V + 0.1782 \cdot V \cdot G$	0.66	1,359.76	3.32	3.49
GPS[CORRECTED]	$2.4037 + 2.5107 \cdot V + 0.2828 \cdot V \cdot G$	0.82	1,261.87	2.19	2.76
<i>Level walking</i>					
MET prediction					
Actual[SPEED]	$1.3378 + 0.5289 \cdot V$	0.65	76.54	0.20	0.44
GPS[SPEED]	$1.2814 + 0.5453 \cdot V$	0.65	69.59	0.19	0.43
GPS[UNCORRECTED]	$1.2872 + 0.5441 \cdot V - 0.0013 \cdot V \cdot G$	0.65	82.67	0.17	0.44
GPS[CORRECTED]	$1.3060 + 0.5382 \cdot V - 0.0080 \cdot V \cdot G$	0.65	77.89	0.17	0.44
$\dot{V}O_2$ (ml $O_2 \cdot \text{min}^{-1} \cdot \text{kg}^{-1}$) prediction					
Actual[SPEED]	$4.6367 + 1.8667 \cdot V$	0.65	294.57	0.69	1.53
GPS[SPEED]	$4.4386 + 1.9242 \cdot V$	0.66	287.44	0.64	1.52
GPS[UNCORRECTED]	$4.4579 + 1.9203 \cdot V - 0.0043 \cdot V \cdot G$	0.66	298.07	0.57	1.53
GPS[CORRECTED]	$4.5229 + 1.9001 \cdot V - 0.0298 \cdot V \cdot G$	0.65	293.49	0.59	1.52

Equations together with statistical parameters (R^2 , AIC, and SEE) obtained with linear mixed models are presented. The accuracy (RMSE) obtained for each model using the leave-one-out method is also displayed. MET, metabolic equivalent; Actual[SPEED AND GRADE], model obtained using the actual speed and grade; GPS[UNCORRECTED], model obtained using the GPS speed and grade, with the grade computed using uncorrected GPS altitude data; GPS[CORRECTED], model obtained using the GPS speed and grade, with the grade computed using corrected GPS altitude data obtained from map projection software; Actual[SPEED], model obtained using the actual speed only; GPS[SPEED], model obtained using the GPS speed only; $\dot{V}O_2$, oxygen uptake; V , speed (km/h); G , grade (%); R^2 , R -squared for linear mixed model; AIC, Akaike information criteria (AIC values are used to compare the goodness of fit of each model: the lower the AIC value, the better the predictive model); SEE, standard error of estimate; RMSE, root-mean-square error obtained with the leave-one-out cross-validation methodology.

values or GPS parameters for speed and grade, for the combined level and uphill walking conditions (Fig. 4) and for the level walking condition only (Fig. 5).

DISCUSSION

To the best of our knowledge, this is the first study to determine the accuracy of using a low-cost GPS receiver to

estimate the metabolic rate during level and uphill outdoor walking; specifically, this study characterized the direct relationship between the speed and grade measured with a GPS receiver and the actual metabolic demand while walking under different controlled speed and grade conditions. As a complementary result, we provide the first speed- and grade-based equations for estimating metabolic rate during level and uphill walking from metabolic data that were exclusively acquired in an outdoor setting.

Accuracy of GPS for Estimating Outdoor Walking Metabolic Rate

Considering the methodological conditions implemented in the present study, the results reported here are unique. In the study by Demczuk (13), which aimed to assess the EE of soldiers during outdoor patrols, the GPS receiver and the indirect calorimeter were not worn at the same time by the same subjects. This could explain the wide ranges in the correlation coefficients (0.34–0.93) and related errors (–273 to 49 J/s; i.e., –784 to 141 ml O_2 /min), between the predicted and actual EE values that the authors reported (13). Hongu et al. (22) studied the accuracy of GPS watches for predicting the EE at different walking speeds, but only during level walking; furthermore, a hip-worn accelerometer was used as the reference method for predicting the actual EE. The studies by Duncan et al. (14) and Nguyen et al. (33) provide interesting data on the use of a GPS receiver to estimate the EE during level and graded outdoor walking. Duncan et al. (14) showed that during a 4-km (2.5-mile) walk, the EE estimation improved by 11% when the grade computed using the GPS data was used compared with the speed results provided by accelerometry alone (14). In the study by Nguyen et al. (33), which used a GPS receiver and two accelerometers (worn at the hip and the ankle), the EE was estimated by including the GPS speed and grade in a published prediction equation (3). However, no conclusion regarding the accuracy of using a GPS receiver to estimate the EE during level and graded walking outdoors could be drawn from the Duncan et al. and Nguyen et al. studies. This is because the direct relationship between the GPS parameters (speed and grade) and the actual metabolic demand while walking under different controlled conditions of speed and grade was not studied.

Our study is the first to characterize this direct relationship during level and graded outdoor walking and using a GPS receiver despite highly challenging experimental constraints. Interestingly, the range of walking speeds and positive grades tested in the present study encompassed most of those that are encountered in real-life outdoor settings. Because finding outdoor pathways that include both different grades and sufficient

Table 3. Cross-validation of published prediction equations for level and uphill walking metabolic rates ($\dot{V}O_2$ in ml $O_2 \cdot \text{min}^{-1} \cdot \text{kg}^{-1}$)

Equations	Actual Speed and Grade ($n = 270$)				GPS Speed and Uncorrected GPS Grade ($n = 267$)				GPS Speed and Corrected GPS Grade ($n = 267$)			
	Bias (95% CI)	SD	95% LoA	RMSE	Bias (95% CI)	SD	95% LoA	RMSE	Bias (95% CI)	SD	95% LoA	RMSE
ACSM (2)	–1.76 (–2.10; –1.42)	2.83	–7.31; 3.79	3.33	–2.07 (–2.56; –1.57)	4.10	–10.10; 5.97	4.59	–1.66 (–2.06; –1.27)	3.29	–8.12; 4.79	3.68
Pandolf et al. (35)	–2.81 (–3.14; –2.48)	2.77	–8.24; 2.62	3.94	–3.13 (–3.62; –2.65)	4.01	–10.99; 4.73	5.08	–2.75 (–3.11; –2.38)	3.04	–8.72; 3.22	4.10

$\dot{V}O_2$, oxygen uptake; n , number of walking periods used for the cross-validation; 95% IC, 95% confidence interval; SD, SD of the differences between the measured and the predicted values; LoA, limits of agreement; RMSE, root-mean-square error. Bias is the mean of the differences between the predicted and the measured metabolic rate values.

Table 4. Cross-validation of published prediction equations for level walking metabolic rate ($\dot{V}O_2$ in ml $O_2 \cdot \text{min}^{-1} \cdot \text{kg}^{-1}$)

Prediction Equations	Actual Speed ($n = 90$)				GPS Speed ($n = 90$)			
	Bias (95% CI)	SD	95% LoA	RMSE	Bias (95% CI)	SD	95% LoA	RMSE
Ludlow and Weyand[ADULTS] (28)	-0.32 (-0.68; 0.03)	1.72	-3.69; 3.04	1.74	-0.37 (-0.73; -0.01)	1.73	-3.75; 3.02	1.75
Ludlow and Weyand[GENERAL] (28)	-0.09 (-0.46; 0.28)	1.77	-3.55; 3.38	1.76	-0.13 (-0.50; 0.24)	1.78	-3.62; 3.36	1.78
Ludlow and Weyand[DOUGLAS] (28)	-0.19 (-0.56; 0.18)	1.77	-3.65; 3.28	1.77	-0.23 (-0.60; 0.14)	1.78	-3.72; 3.26	1.79
Blessey et al. (6)	-0.51 (-0.87; -0.15)	1.73	-3.90; 2.88	1.79	-0.54 (-0.91; -0.18)	1.75	-3.97; 2.88	1.82
Van der Walt and Wyndham (47)	-0.13 (-0.53; 0.27)	1.90	-3.85; 3.59	1.89	-0.19 (-0.58; 0.21)	1.88	-3.86; 3.49	1.87
ACSM (2)	-1.82 (-2.19; -1.46)	1.73	-5.22; 1.57	2.51	-1.84 (-2.20; -1.47)	1.74	-5.25; 1.57	2.52
Pearce et al. (36)	-1.99 (-2.34; -1.63)	1.69	-5.31; 1.33	2.61	-2.04 (-2.39; -1.68)	1.71	-5.38; 1.31	2.65
Balogun et al. (4)	-0.72 (-1.28; -0.16)	2.69	-6.00; 4.56	2.77	-0.79 (-1.35; -0.22)	2.69	-6.06; 4.49	2.79
Pandolf et al. (35)	-2.28 (-2.66; -1.91)	1.80	-5.81; 1.25	2.90	-2.34 (-2.71; -1.96)	1.79	-5.84; 1.17	2.94
Bubb et al. (9)	0.44 (-0.20; 1.08)	3.06	-5.57; 6.45	3.08	0.37 (-0.27; 1.01)	3.05	-5.60; 6.34	3.05
Brooks et al. (8)	-2.31 (-2.78; -1.83)	2.27	-6.75; 2.14	3.23	-2.34 (-2.80; -1.87)	2.22	-6.88; 2.01	3.21

GPS, global positioning system; n , number of walking periods used for the cross-validation; 95% CI, 95% confidence interval; SD, SD of the differences between the predicted and the measured values; LoA, limits of agreement; RMSE, root-mean-square error. Bias is the mean of the differences between the predicted and the measured metabolic rate values.

distances to achieve a steady state at the different walking speeds tested in the present study was highly challenging, we could not test walking speeds higher than 5 km/h. Such faster speeds can often be chosen by people when walking briskly on the level (31). However, because of the high metabolic rate that could be achieved, it is expected that walking at a pace higher than 5 km/h, when the grade is 3% and most of all 10%, may occur less frequently. Our primary finding is that the use of GPS speed and grade parameters allowed the accurate prediction of metabolic rate during level and uphill outdoor walking, particularly when the GPS grade was corrected. Accurate prediction was also obtained with the computed models that used the GPS speed for the level walking condition only (Table 2). Interestingly, including the GPS grade in the model for the level walking condition only did not decrease the accuracy, regardless of whether the uncorrected or corrected GPS grade was used (Table 2).

Since the middle of the 20th century, a number of studies have focused on predicting EE during human walking using speed and grade (Table 1), leading to very popular equations that are widely used in the field of exercise physiology (2, 35). The accuracy we obtained in the present work seems to be within the ranges of these previous studies. However, the direct comparison of our results with these studies remains somewhat difficult because 1) despite the series of important works

provided by Givoni and Goldman (18), Pandolf et al. (35), and Soule and Goldman (42) for the estimation of energy expenditure during level and graded walking, to our knowledge no study proposed equations for both level and uphill walking from data acquired during outdoor experiments only; 2) most of the time equations obtained were not cross-validated as required (43).

Interestingly, our results compared well with previous studies, which used other sensors and other parameters to predict EE but also used mixed modeling and/or leave-one-out approaches (7, 20). Brage et al. (7), in a population with physical characteristics that were relatively similar to those of our sample (in terms of height, weight, and body mass index; however, our population was younger and fitter), studied the accuracy of combining accelerometry and heart rate monitoring to estimate the metabolic rate during treadmill walking. Using the simplest individual calibration procedure, the authors obtained an average RMSE of 53 J·min⁻¹·kg⁻¹ (i.e., 2.54 ml $O_2 \cdot \text{min}^{-1} \cdot \text{kg}^{-1}$) and 87 J·min⁻¹·kg⁻¹ (i.e., 4.16 ml $O_2 \cdot \text{min}^{-1} \cdot \text{kg}^{-1}$) for predicting the metabolic rate during level treadmill walking (speed range 3.2–5.2 km/h) and graded treadmill walking (grade range 1.7–10.2%, with speed defined as “brisk walking,” i.e., between 5.2 and 5.8 km/h), respectively. More recently, Hall et al. (20) found an average RMSE of 1.51 METs for an equation that was developed to predict the

Table 5. Comparison of the measured metabolic rate values with the predicted values obtained using the equations for level walking metabolic rate ($\dot{V}O_2$ in ml $O_2 \cdot \text{min}^{-1} \cdot \text{kg}^{-1}$) and either actual speed or GPS speed

	Measured	Predicted Using Actual Speed	Predicted Using GPS Speed
Present study	10.7 (8.8–13.4)		
Ludlow and Weyand[ADULTS] (28)		10.4 (8.3–13.7)	10.3 (8.3–13.5)
Ludlow and Weyand[GENERAL] (28)		10.7 (8.5–13.9)	10.6 (8.6–13.7)
Ludlow and Weyand[DOUGLAS] (28)		10.6 (8.5–13.8)	10.5 (8.5–13.5)
Blessey et al. (6)		10.3 (8.5–13.2)	10.3 (8.6–13.1)
Van der Walt and Wyndham (47)		10.5 (7.5–15.1)	10.4 (7.6–15.0)
ACSM (2)		9.3 (6.8–11.8)*	9.3 (7.0–11.8)*
Pearce et al. (36)		8.7 (6.6–12.2)*	8.6 (6.7–12.1)*
Balogun et al. (4)		10.7 (9.3–11.3)	10.6 (9.3–11.2)*†
Pandolf et al. (35)		8.4 (5.6–12.7)*	8.3 (5.7–12.5)*
Bubb et al. (9)		12.0 (10.4–12.4)	11.9 (10.4–12.2)
Brooks et al. (8)		8.9 (4.8–13.0)*	8.8 (5.0–12.7)*

Values are medians (IQR). $\dot{V}O_2$, oxygen uptake. * $P < 0.05$ vs. the measured metabolic rate values; † $P < 0.05$ vs. the predicted metabolic rate values using actual speed.

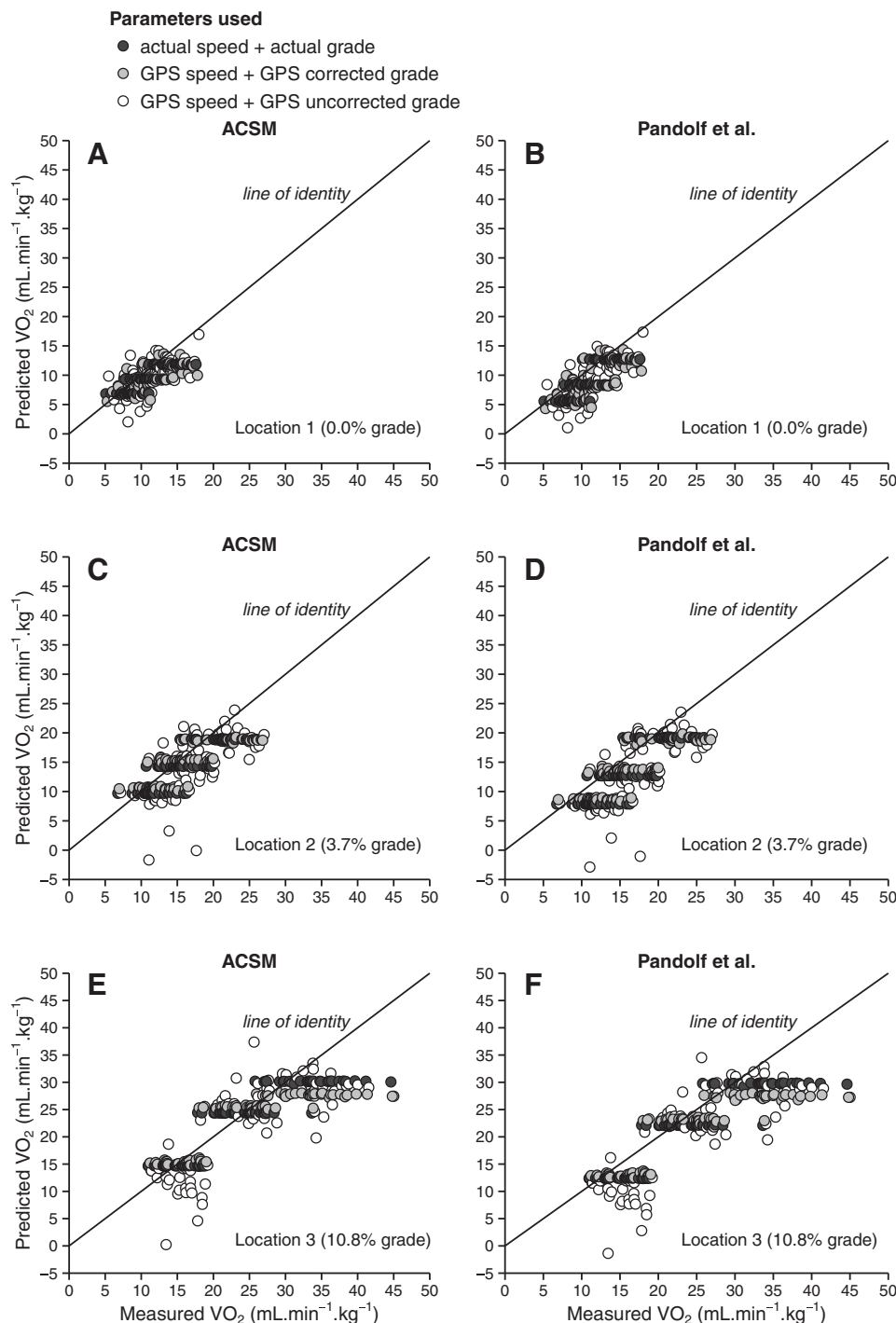


Fig. 4. Gross walking metabolic rates predicted by two standardized equations vs. the metabolic rate values measured during the level and uphill walking sessions: the ACSM (2) equation (A, C, and E) and the Pandolf et al. (35) equation (B, D, and F). In A–F, the model-predicted values obtained using the actual speed and grade, the GPS speed and GPS grade corrected using map projection software, and the GPS speed and uncorrected grade appear in black, grey, and white, respectively. The model-predicted values were obtained for level outdoor walking (A and B), for uphill outdoor walking on a 3.7% mean grade (C and D), and for uphill outdoor walking on a 10.8% mean grade (E and F).

metabolic rate in older adults using accelerometer counts during level walking.

The cross-validation of published speed- and speed/grade-based equations showed that the metabolic rate estimated with GPS speed and grade parameters was very close to the value reached using the actual speed and grade (Tables 3–5 and Figs. 4 and 5). This finding indicates that GPS speed and grade parameters can be implemented with confidence when using the available equations to predict metabolic rate during outdoor walking. Interestingly, our cross-validation results indicate the accuracy of the different equations used (Tables 3–5). For the

level outdoor walking condition only (Tables 4 and 5), the use of the generalized equation recently developed by Ludlow and Weyand (28) for adults yielded a very accurate prediction of the metabolic rate. This underlines the interest in developing the most general and accurate equations possible for predicting metabolic rate, specifically ones that could be used with GPS receivers in free-living conditions. Because the equations of Ludlow and Weyand (28) were developed for level walking only, general (if possible) and accurate predictive equations for both level and graded walking are required. For the combined level and uphill outdoor walking conditions, the results ob-

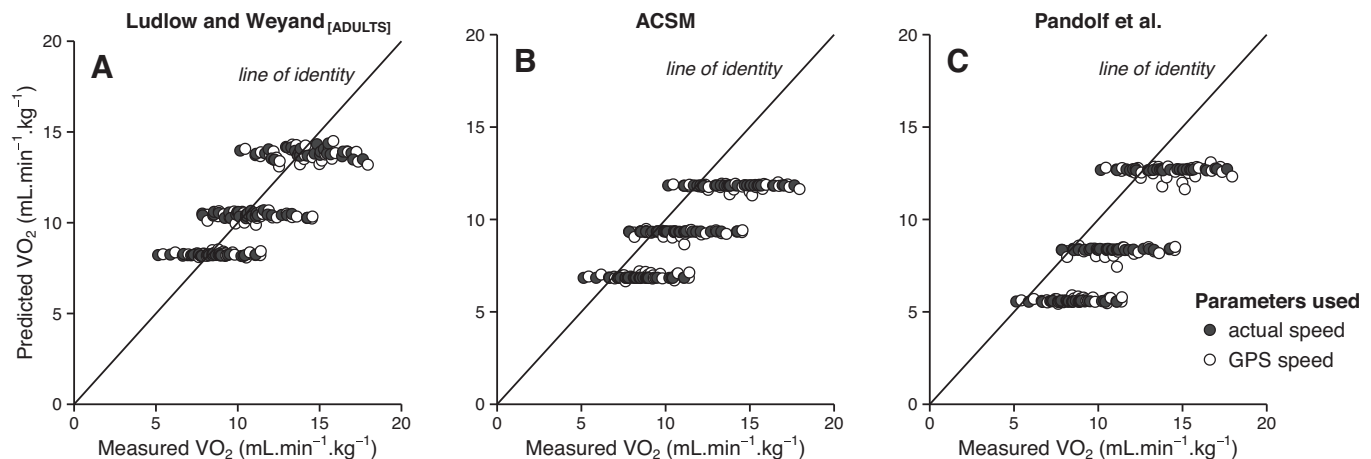


Fig. 5. Gross walking metabolic rates predicted by three equations vs. the metabolic rate values measured during the level walking sessions only: the Ludlow and Weyand (28) equation developed for adults only (A), the ACSM (2) equation (B), and the Pandolf et al. (35) equation (C). In A–C, the model-predicted values obtained using the actual speed and the GPS speed appear in black and white, respectively.

tained using the ACSM (2) equation seemed more accurate than those obtained using the Pandolf et al. (35) equation (Table 3). Importantly, this last result may have been influenced by the range of walking speeds we used. In fact, some studies showed that at faster walking speeds (in particular, beyond 5.5 km/h or 1.5 m/s), the Pandolf et al. equation outperforms the ACSM equation for level walking (28, 49).

Impact of GPS Speed and Grade Measurement Compared with the Use of Actual Speed and Grade for Estimating Outdoor Walking Metabolic Rate

Our results showed that the accuracy obtained for the models we computed using both the GPS speed and corrected grade was very close to the accuracy obtained for the models computed using both the actual speed and grade. However, although it remained satisfactory, the accuracy of the prediction model obtained using GPS parameters of speed and uncorrected grade was decreased (Table 2).

The decrease in accuracy that we observed for predicting the metabolic rate when using the GPS speed and uncorrected grade was mainly explained by the error caused by the GPS receiver's measurements of altitude changes, which led to imprecise grade calculation. Indeed, as shown in Fig. 1, the error in speed estimation was quite low, resulting in a small impact, if any, on metabolic rate estimation. Our results emphasize the value of using map projection software when available to correct the altitude data changes obtained with low-cost GPS receivers. The method we used to correct altitude data is low-cost, rapid, and simple; it only requires the use of an automatic function available on low-cost cartography software. It is likely that the results obtained from the corrected GPS grade could differ according to the map projection software used. Indeed, the methods for computing actual altitude can differ among the different map projection software used, leading to different levels of accuracy.

Our results also show that the GPS grade estimation errors decreased with increases in walking speed (Fig. 2). This result is explained primarily by the methodology used to calculate the grade [(altitude variation/traveled distance) $\times$ 100] and because the distance walked increased with increased walking

speed over a given walking duration. Because the error in altitude variation estimation seemed nearly constant across walking speeds (data not shown, available on simple request), it is logical to determine that greater travel distances over a fixed recording epoch would decrease the error of grade estimation. Finally, as Fig. 2 shows, the GPS receiver's absolute error for grade estimations obtained with corrected altitude data tended to be quite constant across the different grade levels. Consequently, the GPS receiver's relative error for the estimation of metabolic rate while walking increased as the grade approached a value of 0% (data not shown).

Study Limitations

The accuracy of using GPS receiver measurements to estimate the speed of displacement and position is affected by a number of well-known factors that have been previously addressed (39), including the number and geometrical arrangement of available satellites, atmospheric effects, and the level of environmental obstruction created by buildings, dense vegetation, or topographical features. Except for environmental obstructions, these factors were not and could not be controlled in the present study; the same is true for other studies that use GPS data to track physical activity and behavior in free-living conditions. This limit may preclude the generalization of our results to environments other than those used in the present study. Available studies show that dynamic accuracy in latitude/longitude determination was decreased with increase in the level of environmental obstruction (5, 39, 40). The same is probably true for altitude estimation. Thus it could be worthwhile to determine the impact of the obstruction level on the accuracy of EE estimations based on speed and grade data provided by a GPS receiver. However, on a day-to-day basis, we believe that environmental obstruction should have only a moderate impact on the estimation of the metabolic rate during outdoor walking if the GPS's altitude is corrected via map projection software, as proposed here. Furthermore, some GPS devices allow the dilution of precision to be calculated (39, 40), and this could provide confidence ratings for the results obtained during free-living monitoring.

We used only one brand of low-cost GPS receiver in the present study. Although a few studies have compared the accuracy of different GPS receivers for estimating walking speed (34), inter-GPS receiver model variations appear to be low (1, 26, 34) and likely have a small or undetectable effect on the estimation of the metabolic rate during walking. No published data are available regarding altitude estimation using available low-cost GPS receivers. Finally, rather than using both a low-cost GPS receiver and map projection software to correct altitude data, the use of a GPS in Real-Time Kinematic mode (45) could have improved grade estimation and thus EE estimation while walking. While such approaches offer very interesting possibilities, such highly accurate GPS devices remain unsuitable for large samples and free-living applications because of their high cost, the cumbersome nature of the equipment, and the complexity of the posttreatment data (45).

Because the present study focused only on outdoor walking, our results cannot be extended to the measurement of other physical activities that can be performed under real-life conditions. However, we decided to focus on walking only because it is a physical activity that is commonly reported in the general population (15, 41) and is recommended by scientific authorities for both health (21) and rehabilitation (16) purposes. Furthermore, our study assessed walking outdoors only and not indoors because the GPS method can only be accurately used when there is a clear line of sight to the satellites. While this could be a limitation for assessing physical activity patterns continuously in free-living conditions, this limitation may not be significant for assessing walking behavior from epidemiological and interventional standpoints. In fact, while physical activity guidelines recommend achieving the minimum amount of moderate physical activity via accumulated bouts of 10 min or more (21), it may be that such walking durations are more easily achieved outdoors (as during active transport) than indoors, except when walking exercise is performed on a treadmill.

Another potential limitation of the present study is that it was difficult to determine the exact effect of the terrain surface on the $\dot{V}O_2$ value during the graded outdoor walking sessions. We estimated coefficients on the basis of previously published data (42) and used them to adjust the $\dot{V}O_2$ data according to the effect of the surface on the energy cost of walking. Interestingly, when we developed linear mixed models using unadjusted metabolic data, although the $\dot{V}O_2$ values and the equations were modified, this had no effect on the comparison of the statistical models computed using either actual or GPS speed and grade parameters (data not shown). Further, we found that the RMSEs reached for the models developed using adjusted metabolic data were lower (i.e., a lower error, so a better estimation), compared with the models developed using unadjusted metabolic data (data not shown).

Conclusion

Using a linear mixed modeling approach, we show that a low-cost GPS receiver provides an accurate estimation of the metabolic rate during level and graded outdoor walking. This estimation is improved when the GPS receiver is used with map projection software to correct the altitude data and thus the grade. Importantly, we also show that the accuracy obtained using previously published predictive equations for the walk-

ing metabolic rate is not affected by the use of GPS parameters (in particular, the use of GPS speed and corrected grade) compared with the use of the actual speed and grade. These findings provide new perspectives for the study of human EE while walking in free-living conditions and offer promising health and disease applications.

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DISCLOSURES

No conflicts of interest, financial or otherwise, are declared by the author(s).

AUTHOR CONTRIBUTIONS

P.-Y.d.M., M.G., G.M., S.B., B.N.-D., and A.L.F. conception and design of research; P.-Y.d.M., R.D., M.G., G.M., S.B., and A.L.F. performed experiments; P.-Y.d.M., R.D., A.L., and A.L.F. analyzed data; P.-Y.d.M. and A.L.F. interpreted results of experiments; P.-Y.d.M. and A.L.F. prepared figures; P.-Y.d.M. and A.L.F. drafted manuscript; P.-Y.d.M., R.D., M.G., G.M., A.L., S.B., J.P., B.N.-D., and A.L.F. edited and revised manuscript; P.-Y.d.M., R.D., M.G., G.M., A.L., S.B., J.P., B.N.-D., and A.L.F. approved final version of manuscript.

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